

Improving Supply Chain Resilience under Partial Visibility: A Computational Study of Monitoring and Backup Sourcing Policies

Independent Research Project

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Abstract

This report examines how a focal firm can improve supply chain resilience when it observes only part of a multi-tier supplier network and faces a limited resilience budget. I develop a stylized computational model in which the firm can allocate resources to two local interventions: expanding upstream visibility and assigning backup sourcing for selected tier-1 suppliers. The framework is designed to distinguish between two resilience mechanisms: information acquisition through monitoring and local redundancy through backup sourcing. The analysis compares targeted and random policies across three network topologies: sparse, dense, and hub-dominated networks. The results show that the effectiveness of resilience investments depends strongly on network structure. In sparse networks, targeted backup sourcing performs best, while in dense and hub-dominated networks targeted visibility produces the largest reduction in disruption exposure. Across topologies, structurally targeted policies outperform random allocation of the same resources. These findings suggest that even under partial visibility, firms can materially improve resilience through well-targeted local actions, and that the value of those actions depends on how disruption risk is organized in the network.

1. Introduction

Modern supply chains are deep, interconnected, and only partially visible to the firms operating within them. A focal firm typically knows its direct tier-1 suppliers but has limited information about upstream tiers where many disruptions originate. This creates a managerial problem: how should a firm allocate a limited resilience budget when it does not fully observe the network that generates its disruption risk?

This report develops a stylized computational framework for that question. A focal firm faces upstream disruptions in a multi-tier supplier network and can respond through two local decisions. First, it can invest in visibility by monitoring selected upstream nodes. Second, it can protect selected tier-1 suppliers through backup sourcing. These choices are evaluated under different network structures to study whether the value of visibility and backup sourcing depends on topology.

The project is motivated by three research questions. First, how much can a focal firm reduce disruption exposure with a fixed resilience budget? Second, does the relative value of visibility and backup sourcing depend on network topology? Third, do structurally targeted

interventions outperform random allocation of the same resources? More broadly, the report makes three contributions. First, it develops a stylized computational framework linking partial upstream visibility to resilience decisions under a fixed budget. Second, it distinguishes between two resilience mechanisms: information acquisition through monitoring and local redundancy through backup sourcing. Third, it shows that the relative value of these mechanisms depends on network structure, with different policy rankings emerging in sparse, dense, and hub-dominated supply networks.

2. Model and Experimental Design

2.1 Network structure

The supply network is modeled as a directed layered graph with three supplier tiers upstream of a focal firm. Edges run from supplier to customer. Tier-1 suppliers connect directly to the focal firm, tier-2 suppliers feed tier-1 suppliers, and tier-3 suppliers feed tier-2 suppliers. Three stylized topologies are considered: sparse, dense, and hub-dominated.

Sparse networks are characterized by lower branching and weaker overlap across supplier branches. Dense networks contain more shared upstream structure, increasing the possibility that one disruption affects multiple downstream branches. Hub-dominated networks represent concentrated dependence on a smaller set of critical upstream suppliers.

Figure 1 illustrates an example of the synthetic multi-tier (hub) network used in the computational study.

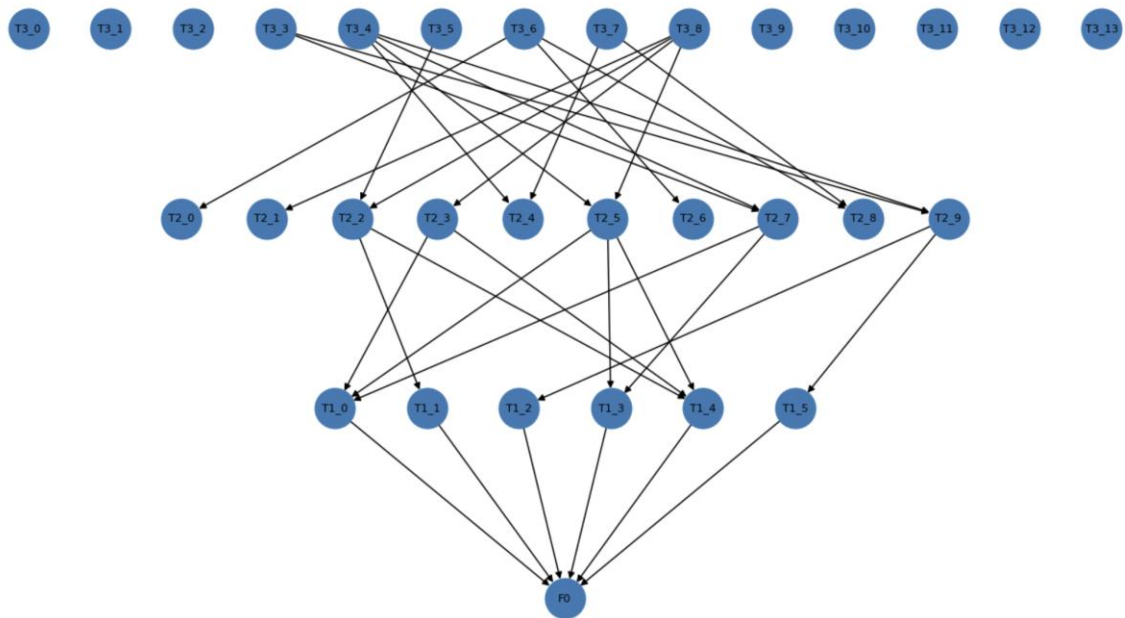


Figure 1. Example of a synthetic multi-tier supply network with three upstream tiers and a focal firm

2.2 Policies

All policies are evaluated under the same total resilience budget. Four main policy classes are compared. The baseline policy makes no additional intervention. The visibility-only policy allocates the full budget to monitoring selected upstream nodes. The backup-only policy

allocates the full budget to backup assignments for selected tier-1 suppliers. The hybrid policy splits the budget between upstream monitoring and backup sourcing. In addition, random visibility, random backup, and random hybrid policies are used as benchmarks to test whether structurally targeted decisions outperform uninformed allocation.

2.3 Targeting rules

Visibility decisions are targeted toward upstream nodes with high downstream tier-1 coverage and high local network importance. Backup assignments are targeted toward tier-1 suppliers with greater visible upstream exposure, subject to a unique-pairing rule in which one backup supplier can protect at most one primary supplier. These targeted policies are compared to random policies with the same budget.

2.4 Disruption process and metrics

In each simulation scenario, upstream nodes may be disrupted with tier-specific probabilities. A tier-1 supplier is considered failed if either the supplier itself is disrupted or at least one relevant upstream node is disrupted, unless the disrupted node is monitored and mitigation succeeds. The main outcome metric is the share of impacted tier-1 suppliers. Additional metrics include the probability of severe disruption and the 90th percentile of disruption impact.

The purpose of the model is not to replicate the full operational complexity of real supply chains, but to isolate how visibility, local redundancy, and network structure interact in a common decision framework.

3. Results

The simulation results show that the effectiveness of resilience investments depends strongly on network topology. More specifically, the analysis highlights three findings: policy effectiveness is topology-dependent, targeted interventions outperform random allocation, and the relative value of visibility and backup sourcing can be understood through different underlying resilience mechanisms.

Figure 2 summarizes the average disruption impact across policies and topologies.

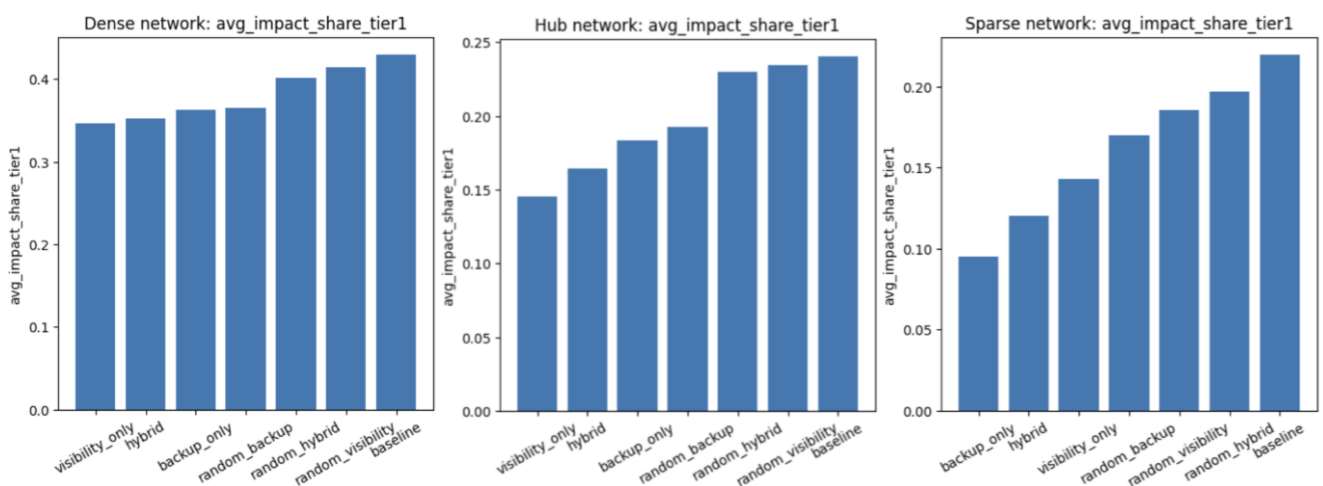


Figure 2. Average disruption impact across policies and network topologies. Targeted policies reduce the share of impacted tier-1 suppliers relative to the baseline and generally outperform random allocation of the same resilience resources

Finding 1. Policy effectiveness depends on topology.

The results show that different network structures favor different resilience strategies. In the sparse topology, the baseline average impact share is 0.220. The best targeted policy is backup only, which reduces the average impact to 0.095. Visibility only and hybrid policies also improve performance, but not as strongly as selective backup sourcing. This pattern is consistent with the idea that when supplier branches are relatively separate, local redundancy can effectively isolate disruption exposure.

In the dense topology, the baseline average impact share is 0.429. The best targeted policy is visibility only, which reduces average impact to 0.338. Backup only and hybrid also outperform the baseline, but to a smaller extent. Here the likely mechanism is different: when multiple downstream branches share upstream structure, disruptions propagate through common nodes, so identifying and monitoring those shared dependencies becomes more valuable than protecting only a small number of tier-1 suppliers.

Figure 3 reports the probability of severe disruption across policies and topologies.

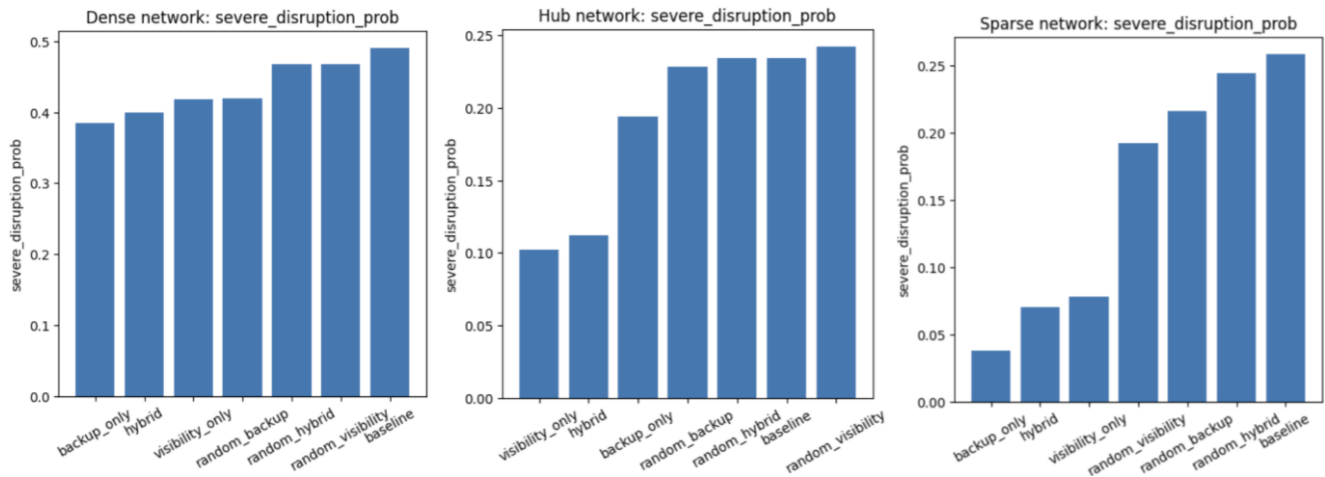


Figure 3. Probability of severe disruption across policies and network topologies. The results indicate that targeted interventions reduce not only average exposure but also the likelihood of large disruption events

In the hub topology, the baseline average impact share is 0.240, while visibility only reduces it to 0.144 and hybrid reduces it to 0.158. Backup only also improves on the baseline, but less strongly. This result again points to the value of information in networks with concentrated dependence. When disruption exposure is driven by a smaller set of critical upstream nodes, monitoring those nodes yields especially high returns.

Finding 2. Targeted policies outperform random allocation.

A second important result is that structurally targeted interventions generally outperform random allocation of the same resilience resources. In the dense topology, targeted visibility achieves an average impact of 0.338, compared with 0.385 for random visibility. In the sparse topology, targeted backup sourcing reduces average impact to 0.095, compared with 0.170 for random backup allocation. Similar patterns appear in the hub topology, where targeted

visibility also outperforms its random counterpart. These differences indicate that resilience investments become more effective when they are informed by network structure rather than assigned without regard to upstream dependence.

Finding 3. Visibility and backup sourcing operate through different mechanisms.

Taken together, the results suggest that the two interventions create resilience through different channels. Visibility primarily improves information quality: it helps the focal firm identify critical upstream dependencies and therefore target monitoring resources where they matter most. Backup sourcing, by contrast, creates local redundancy at the level of tier-1 suppliers. The relative value of these mechanisms depends on how risk is organized in the network. When exposure is concentrated in shared upstream nodes, visibility becomes especially valuable. When exposure is distributed across more separable branches, backup sourcing becomes relatively more effective.

Figure 4 provides a compact summary of policy performance across network structures.

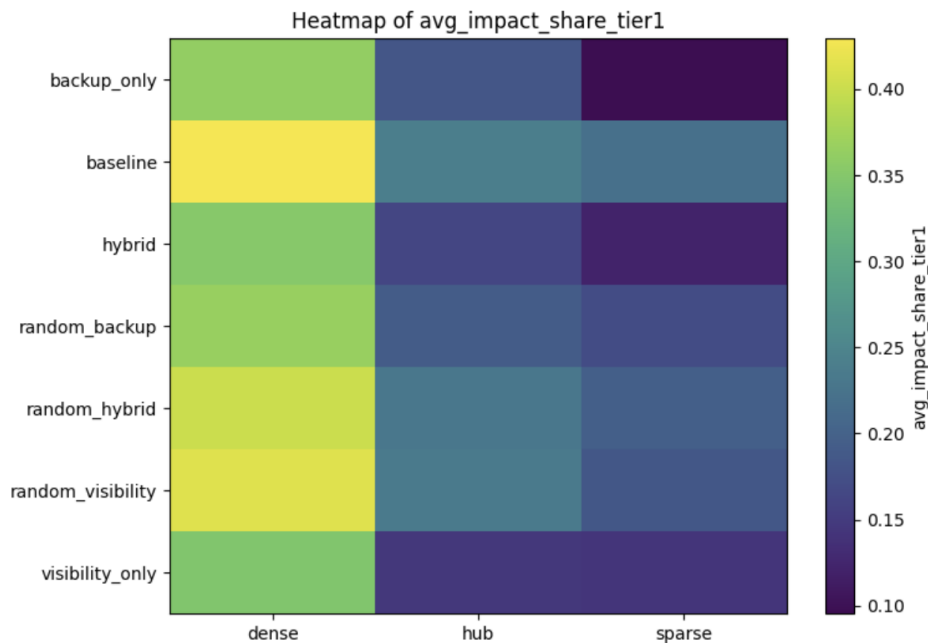


Figure 4. Heatmap of average disruption impact by policy and topology. The relative ranking of policies varies across network structures, indicating that resilience investments are topology-dependent

Overall, the results show that no single resilience policy is uniformly best. Instead, the effectiveness of local interventions depends on the interaction between limited visibility, budget constraints, and network structure.

4. Discussion

The main implication of this computational study is that resilience under partial visibility should be understood as a topology-dependent decision problem rather than a one-size-fits-all resource allocation problem. A focal firm does not need full visibility over the entire supply network in order to improve resilience, but it does need to align its interventions with the way disruption exposure is structured upstream. This is why different policies perform best in different topologies.

The results also clarify why visibility and backup sourcing should not be treated as interchangeable instruments. In this model, visibility creates value by improving the firm's ability to identify structurally important upstream nodes, while backup sourcing creates value by reducing dependence on specific tier-1 branches. These are distinct resilience mechanisms. Their relative usefulness depends on whether disruptions are driven primarily by shared upstream exposure or by weaker interdependence across branches. This distinction is important because it moves the analysis beyond the simple question of whether a policy "works" and toward the more research-relevant question of why it works in some network structures but not in others.

From a managerial perspective, the findings suggest that firms can improve resilience through local, feasible interventions even when they cannot redesign the entire network. This is relevant in practice because focal firms typically have influence over monitoring decisions and sourcing choices at the local level, but only limited control over the broader upstream network. The results therefore support the idea that meaningful resilience gains can be achieved within a restricted sphere of influence, provided that decisions are informed by network structure.

The study also has methodological implications. The consistent gap between targeted and random policies suggests that the value of resilience investments depends not only on how much budget is available, but also on how well the firm can convert structural information into action. In that sense, partial visibility is not merely a descriptive property of supply chains; it is a core feature of the decision problem itself.

At the same time, the model is intentionally stylized and should be interpreted as a mechanism-based computational study rather than a fully realistic representation of supply-chain operations. The use of synthetic networks means that the analysis does not capture the empirical degree distributions or contractual frictions of real supply networks. The disruption process is simplified, and the value of visibility is represented through probabilistic early-warning mitigation. In addition, backup sourcing is modeled as a binary protection decision and does not include capacity limits, congestion, or substitution frictions. These assumptions help isolate the comparative effects of different resilience mechanisms, but they also limit the realism of the current framework.

These limitations point directly to natural research extensions. Future work could test the same logic on empirical supply-chain network data, incorporate capacity-constrained backup sourcing, model correlated disruptions, or embed the budget-allocation problem in an exact optimization framework. Such extensions would strengthen both the empirical and methodological relevance of the approach. Even in its current form, however, the model demonstrates how partial visibility, local interventions, and network structure can be integrated into a tractable Operations Research framework for studying supply-chain resilience.

In that sense, the present report should be understood as a first computational step toward a richer research agenda on partial visibility, network structure, and resilience-oriented sourcing decisions.

5. Conclusion

This report developed a small computational study of supply chain resilience under partial visibility. The main result is that resilience investments are topology-dependent: backup sourcing performs best in sparse networks, while visibility is most valuable in dense and hub-dominated networks. Across all cases, targeted policies outperform random allocation. Taken together, the findings show that even under partial visibility, a focal firm can materially improve resilience through structurally targeted local interventions.